

Linear systems and elimination

Near-quadratic solution cost at a prescribed backward error

Difficulty: challenging **Importance:** broadly interesting**Status:** Solved

Rating rationale (historical): Challenging reflects the remaining removal of a dimension-dependent logarithm from an established general algorithm; broad impact follows from a condition-independent cost bound for arbitrary linear systems.

Resolution: affirmative, 12 September 2026

Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. [Verified affiliation and submission record.](#)

Theorem 1, proved in Sections 2–6 gives a randomized exact-real algorithm with deterministic worst-case cost $O(n^2\epsilon^{-3})$ and $O(n^2)$ scalar storage. For every original input it always returns a nonzero vector and achieves the original A-only backward error at most $5\epsilon/8$ with probability greater than 0.997. Thus it settles the complete target with $q = 3$, uniformly in dimension, conditioning and right-hand side. [Proof source.](#)

The argument passed a separate [independent Codex AI-agent audit](#), including exact component-test reproduction. This is informal automated review, not external human peer review or formal verification. The supplied draft was prepared with ChatGPT assistance; no Lean verification was performed.

Entrywise randomized rounding and weighted-pattern matrix products cancel the logarithmic iteration overhead in total arithmetic cost. This does not assert dimension-independent black-box convergence, finite-precision stability, bit complexity or practical speed. The original target and prior-source attribution below are retained; the earlier status discussion is historical.

Problem statement

Does there exist a randomized algorithm and absolute constants $C, q > 0$ such that, for every $n \geq 1$, nonsingular $A \in \mathbb{R}^{n \times n}$ with $\|A\|_2 = 1$, $b \neq 0$, and $0 < \epsilon < 1/2$, it returns $x \neq 0$ satisfying

$$\Pr \left\{ \frac{\|Ax - b\|_2}{\|A\|_2 \|x\|_2} \leq \epsilon \right\} \geq 0.99$$

using at most $Cn^2\epsilon^{-q}$ operations? Use an exact-real arithmetic model with scalar arithmetic, square roots, comparisons, and independent standard Gaussian or random-bit draws at unit cost. All input entries may be accessed; the norm normalization is an input promise. The cost constants must be independent of the condition number and of n, b, ϵ . The fraction is normwise backward error with perturbations allowed in A only. A deterministic algorithm meeting the bound would also answer positively.

References and status

M. Dereziński, Y. Nakatsukasa, and E. Rebrova, [Towards Universal Convergence of Backward Error in Linear System Solvers](#) (22 May 2026), §§1, 3.1, 5.2, 7, poses the cost question and specifies the backward-error metric. Corollary 17 gives $O(n^2/\sqrt{\epsilon})$ for PSD systems. The revised paper’s Corollary 25 proves the general-system bound $O(n^2 \log(n/\delta)/\epsilon)$ with failure probability δ ; its §7 asks for convergence independent of dimension. The remaining target here removes the $\log n$ overhead, which cannot be hidden in a constant depending only on ϵ . The model and success probability above make this explicit. Searches on 2026-09-08 for “universal backward error 2026 linear systems”, “MINBERR smoothed analysis”, and “quadratic backward error linear solver” found no resolution of the displayed bound. The earlier version’s lack of a proved general-system rate is no longer current.

Audit update — 2026-09-10

Rechecked [version 2](#), Corollaries 17 and 25 and §7. Positive definite inputs satisfy the displayed cost by Corollary 17; the general bound still contains $\log n$. This warrants partial status and a difficulty change from extreme to challenging for the narrowed remaining gap. Searches for universal backward-error convergence and MINBERR follow-ups found no dimension-independent general bound.